

M E R N Data Flow Diagrams & User Stories

shopEZ — B2C E-Commerce Marketplace (MERN Stack)

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Data Flow Diagrams & User

Stories

1. Introduction

This document presents the Data Flow Diagrams (DFDs) and formal User Stories for the ShopEZ platform. DFDs model the system at two abstraction levels — Context Level (Level 0) and Functional Decomposition (Level 1) — illustrating data flows between actors, processes, and data stores.

2. Level 0 — Context Diagram

Login Credentials, Search Queries, Cart Actions, Order Details
Product Catalogue, Cart State, Order Confirmations
Product Data, Category Data, Order Status Updates
Sales Metrics, Order Lists, Platform Reports
Payment Request
Payment Success / Failure
Image Upload Request
Hosted Image URL
Ø=Üd Customer
Ø=Ý Administrator
Ø=Ü³ Payment Gateway
Ø=Ý¼ Cloudinary CDN
Ø=ÞØ ShopEZ E-commerce System

FIGURE 6.1 — LEVEL 0: CONTEXT DIAGRAM (SYSTEM AS BLACK

BOX)

3. Level 1 — Authentication Sub-Process

email, password, name
Hash password (Bcrypt) Validate credentials User record
JWT Token (HTTP-only Cookie)
Ø=Üd Customer
Process 1.0 — User Authentication
Ø=ŸÄ DS-1: Users Collection (MongoDB)

FIGURE 6.2 — LEVEL 1: AUTHENTICATION PROCESS DFD

4. Level 1 — Shopping & Order Placement Sub-

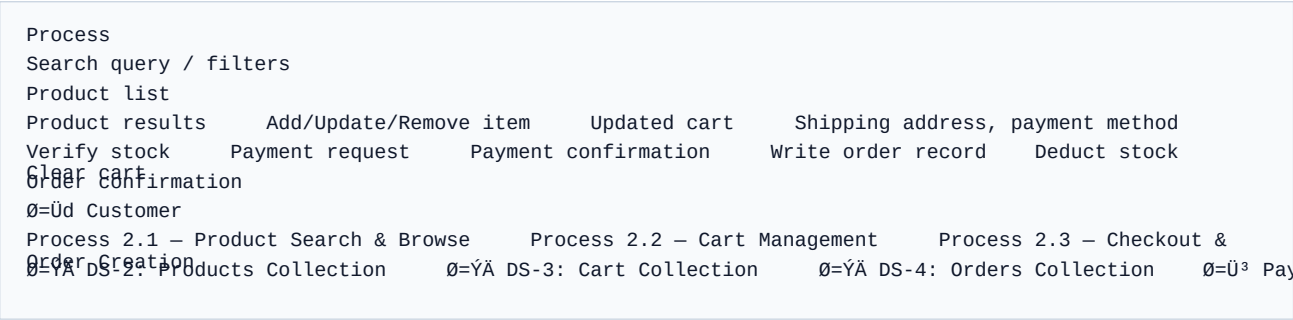


FIGURE 6.3 — LEVEL 1: SHOPPING & ORDER PLACEMENT DFD

5. User Stories

USN	User Type	User Story	Acceptance Criteria	Points	Sprint
USN-1	Customer	As a customer, I can register by entering my name, email, and password.	Account created in DB; redirected to dashboard; duplicate email returns 400.	2	Sprint 1
USN-2	Customer	As a customer, I can log in using my registered email and password.	JWT issued in HTTP-only cookie; incorrect credentials return 401.	2	Sprint 1
USN-3	Customer	As a customer, I can search and filter products by category, price, and rating.	Results accurately reflect filters; update without full page reload.	3	Sprint 1
USN-4	Customer	As a customer, I can add, remove, and update quantities in my cart.	Cart total updates in real time; cart persists across sessions and devices.	3	Sprint 2
USN-5	Customer	As a customer, I can complete a guided checkout by entering shipping and payment details.	All three checkout steps navigable; address validated before proceeding.	3	Sprint 2
USN-6	Customer	As a customer, I can securely pay via an integrated payment gateway.	Payment processed; order status changes to "Paid"; stock deducted; confirmation shown.	5	Sprint 2
USN-7	Customer	As a customer who has completed a purchase, I can leave a star rating and review.	Review appears on the product page; aggregate rating recalculated.	2	Sprint 3
USN-8	Admin	As an admin, I can add, edit, and delete product listings with Cloudinary image uploads.	Changes reflect immediately on storefront; images served via CDN URL.	3	Sprint 1
USN-9	Admin	As an admin, I can view all orders and update their delivery status.	Status transitions: Processing -> Shipped -> Delivered; reflected in customer history.	2	Sprint 3
USN-10	Admin	As an admin, I can view aggregated sales metrics and monthly revenue charts.	Dashboard shows accurate total revenue, order count, and monthly breakdown.	3	Sprint 3

